PARISHRAM 2.0

CLASS 12TH 2026

Biology-

HUMAN REPRODUCTION

ZOOLOGY | LECTURE-2

-PARAM SIR



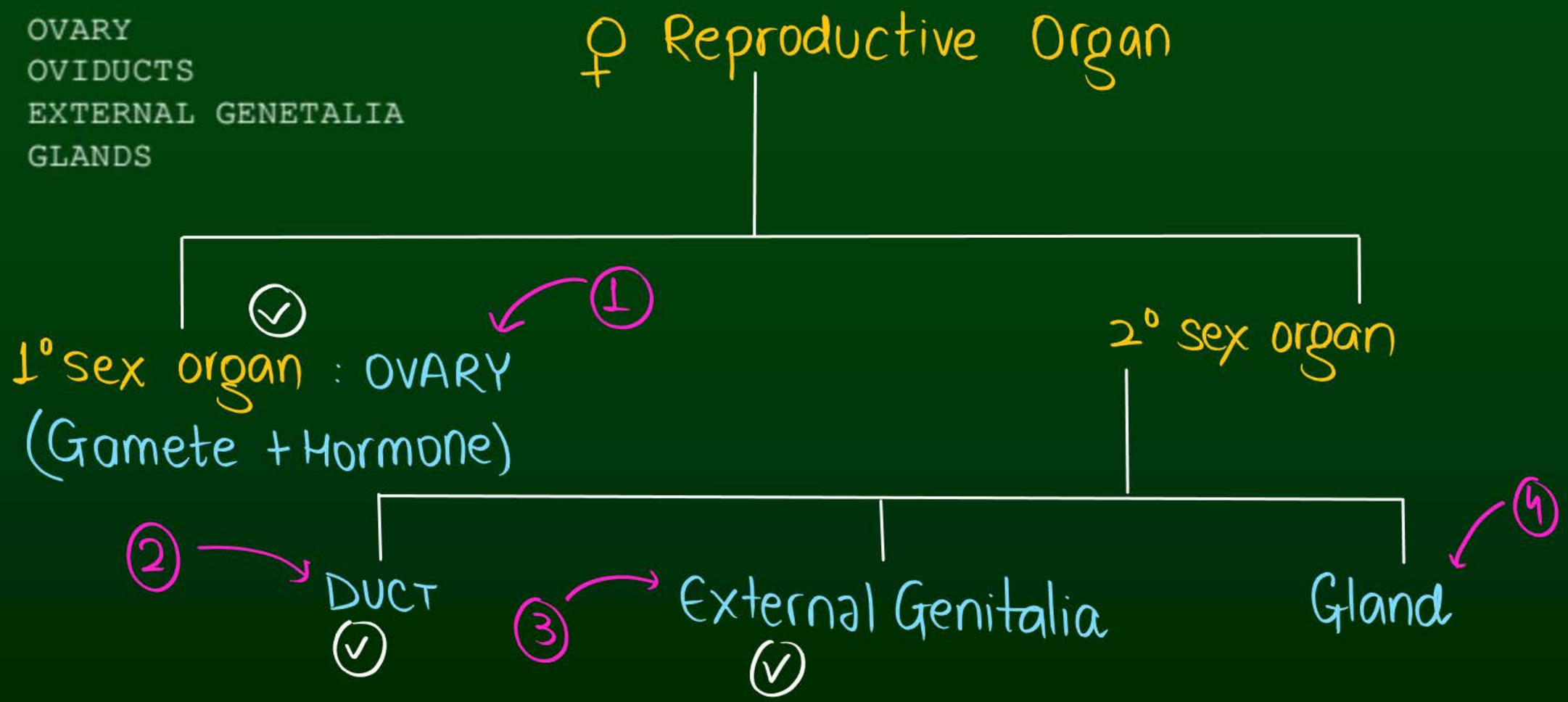
TODAY'S TOPIC -

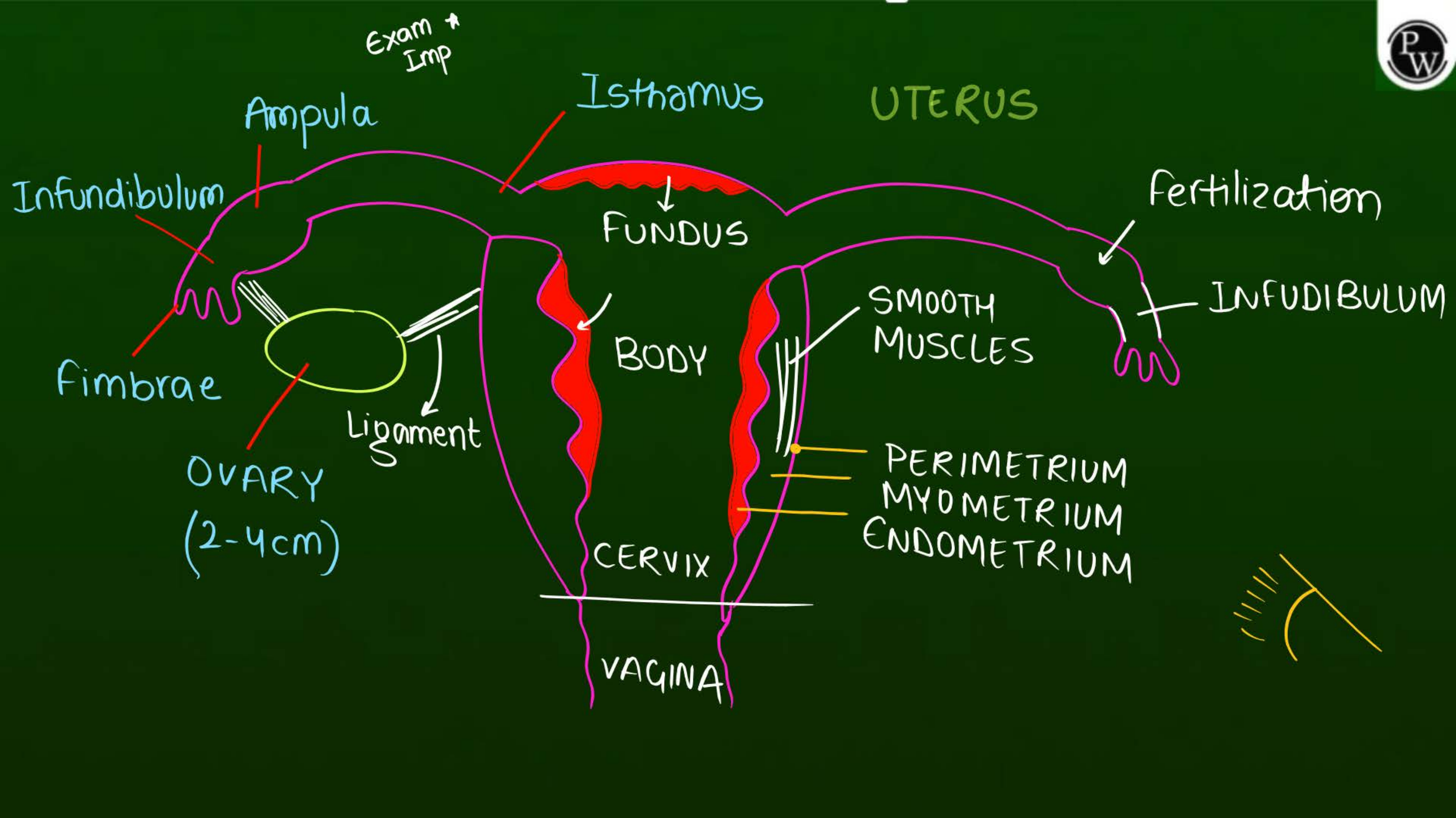
1. REVISION
2. TYPE OF SEX ORGAN IN FEMALE
3. FEMALE REPRODUCTIVE SYSTEM



TYPE OF SEX ORGAN IN FEMALE

- OVARY
- OVIDUCTS
- EXTERNAL GENETALIA
- GLANDS




Exam + Imp
UTERUS
Ampula
Isthamus
Infundibulum
FUNDUS
fertilization
INFUDIBULUM
SMOOTH MUSCLES
BODY
Fimbrae
Ligament
OVARY (2-4cm)
PERIMETRIUM
MYOMETRIUM
ENDOMETRIUM
CERVIX
VAGINA

OVARY

→ Location : Pelvic region

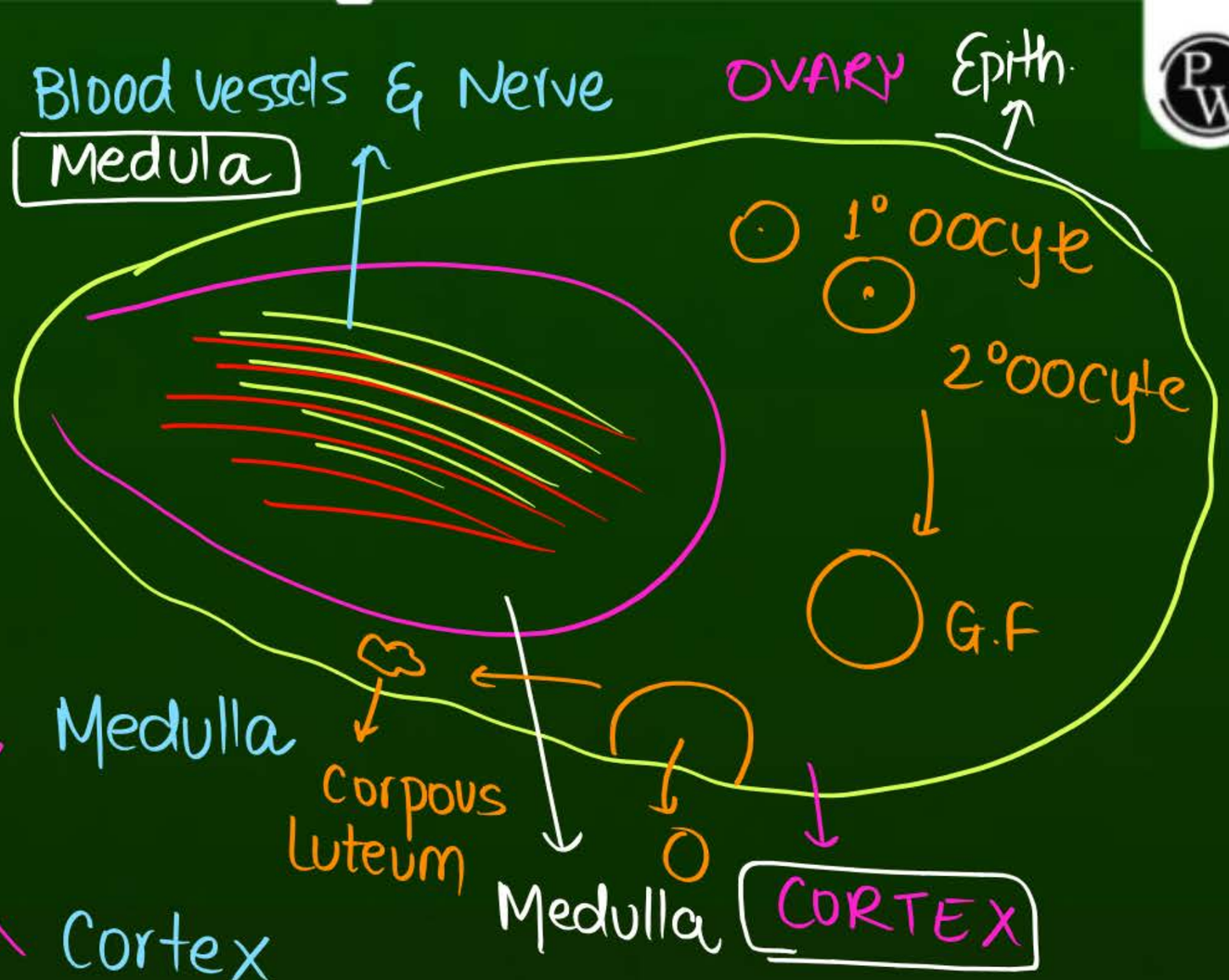
→ Size : 2-4cm

→ Connected by ligment

→ PART

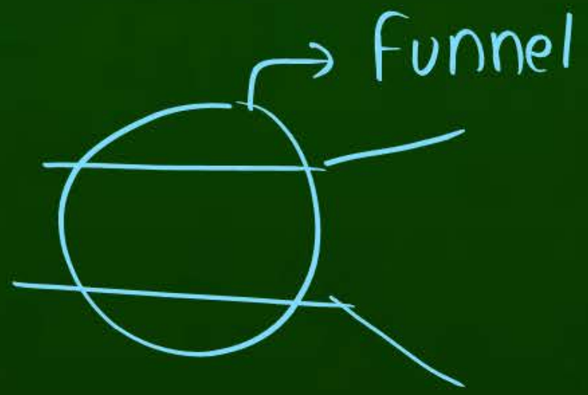
↳ Ovarian Stroma
- Medulla
- Cortex

Cover by epithelium



Fallopian tube (OVIDUCT)

- 10 - 12 cm



Infundibulum

- funnel shape
- It has finger like projection : fimbrae

→ Closest to OVARY

Ampulla *

- Wider
- Fertilization
- Ampullary - isthmic junction

Isthmus

* Close to Uterus

Uterus / Womb : inverted pear shape | single

3 part : fundus, Body, cervix

3 layer : Endometrium : line lumen, Glandular layer, undergo Cyclic Change
Myometrium : Thick layer, Smooth muscle, Contraction
PERIMETRIUM : External thin layer.

→ Uterus open into vagina via cervix

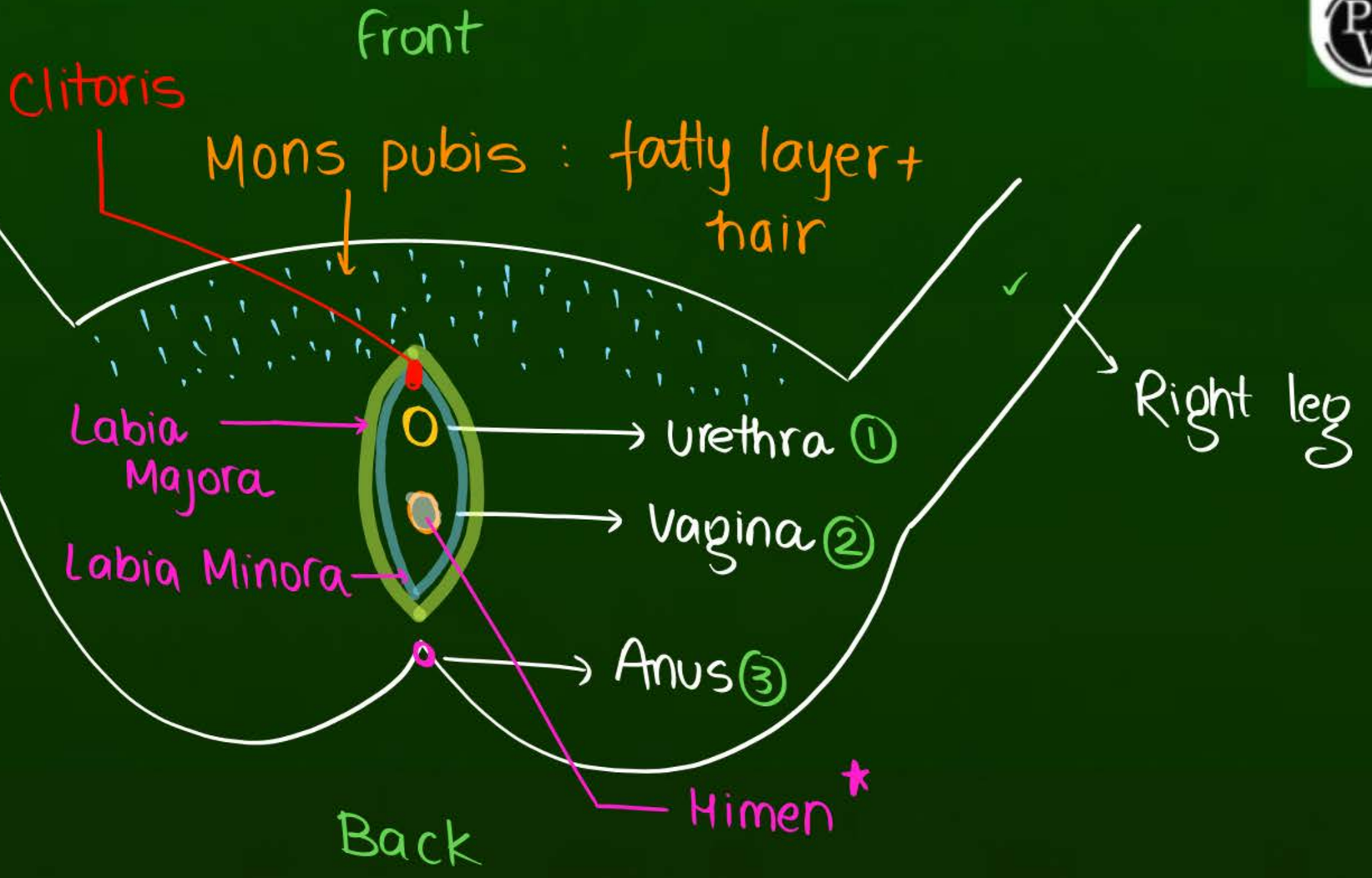
→ Vagina copulatory organ

→ Cervix + vagia = BIRTH CANAL.

External Genitalia

↓

1. Mons pubis
2. labia majora
3. labia Minora
4. Clitoris
5. Hymen



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2.2 The Female Reproductive System

The female reproductive system consists of a pair of **ovaries** alongwith a pair of **oviducts**, **uterus**, **cervix**, **vagina** and the **external genitalia** located in pelvic region (Figure 2.3a). These parts of the system alongwith a pair of the **mammary glands** are integrated structurally and functionally to support the processes of ovulation, fertilisation, pregnancy, birth and child care.

Ovaries are the primary female sex organs that produce the female gamete (ovum) and several steroid hormones (ovarian hormones). The ovaries are located one on each side of the lower abdomen (Figure 2.3b). Each ovary is about 2 to 4 cm in length and is connected to the pelvic wall and uterus by ligaments. Each ovary is covered by a thin epithelium which encloses the ovarian stroma. The stroma is divided into two zones – a peripheral cortex and an inner medulla.

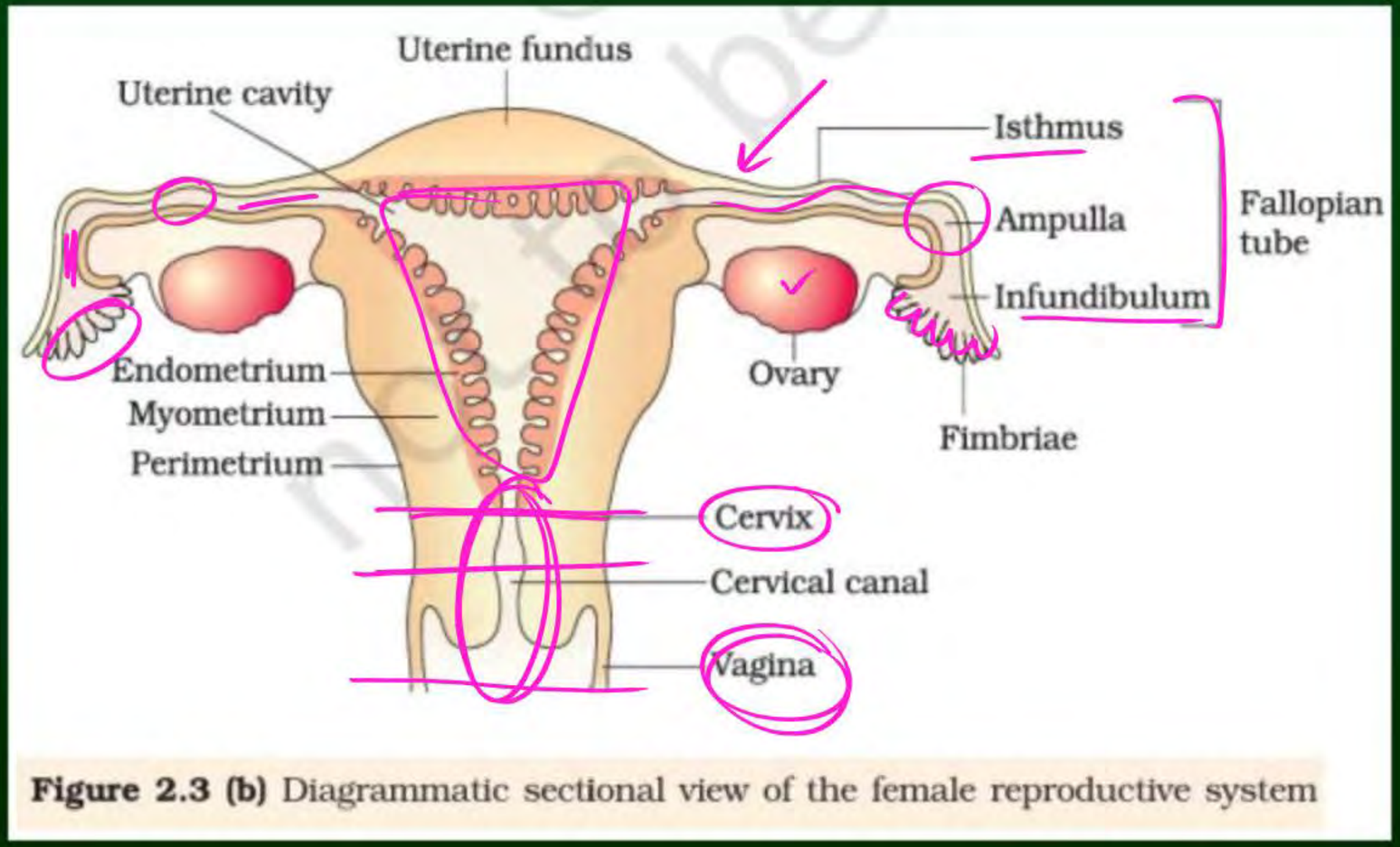


Figure 2.3 (b) Diagrammatic sectional view of the female reproductive system

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The oviducts (fallopian tubes), uterus and vagina constitute the female accessory ducts. Each fallopian tube is about 10-12 cm long and extends from the periphery of each ovary to the uterus (Figure 2.3b), the part closer to the ovary is the funnel-shaped **infundibulum**. The edges of the infundibulum possess finger-like projections called **fimbriae**, which help in collection of the ovum after ovulation. The infundibulum leads to a wider part of the oviduct called **ampulla**. The last part of the oviduct, **isthmus** has a narrow lumen and it joins the uterus.

The uterus is single and it is also called **womb**. The shape of the uterus is like an inverted pear. It is supported by ligaments attached to the pelvic wall. The uterus opens into vagina through a narrow cervix. The cavity of the cervix is called **cervical canal** (Figure 2.3b) which alongwith vagina forms the birth canal. The wall of the uterus has three layers of tissue. The external thin membranous **perimetrium,** middle thick layer of smooth muscle, **myometrium** and inner glandular layer called **endometrium** that lines the uterine cavity. The endometrium undergoes cyclical changes during menstrual cycle while the myometrium exhibits strong contraction during delivery of the baby.

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The female external genitalia include mons pubis, labia majora, labia minora, hymen and clitoris (Figure 2.3a). **Mons pubis** is a cushion of fatty tissue covered by skin and pubic hair. The **labia majora** are fleshy folds of tissue, which extend down from the mons pubis and surround the vaginal opening. The **labia minora** are paired folds of tissue under the labia majora. The opening of the vagina is often covered partially by a membrane called **hymen**. The **clitoris** is a tiny finger-like structure which lies at the upper junction of the two labia minora above the urethral opening. The hymen is often torn during the first coitus (intercourse). *However, it can also be broken by a sudden fall or jolt, insertion of a vaginal tampon, active participation in some sports like horseback riding, cycling, etc. In some women the hymen persists even after coitus. In fact, the presence or absence of hymen is not a reliable indicator of virginity or sexual experience.*

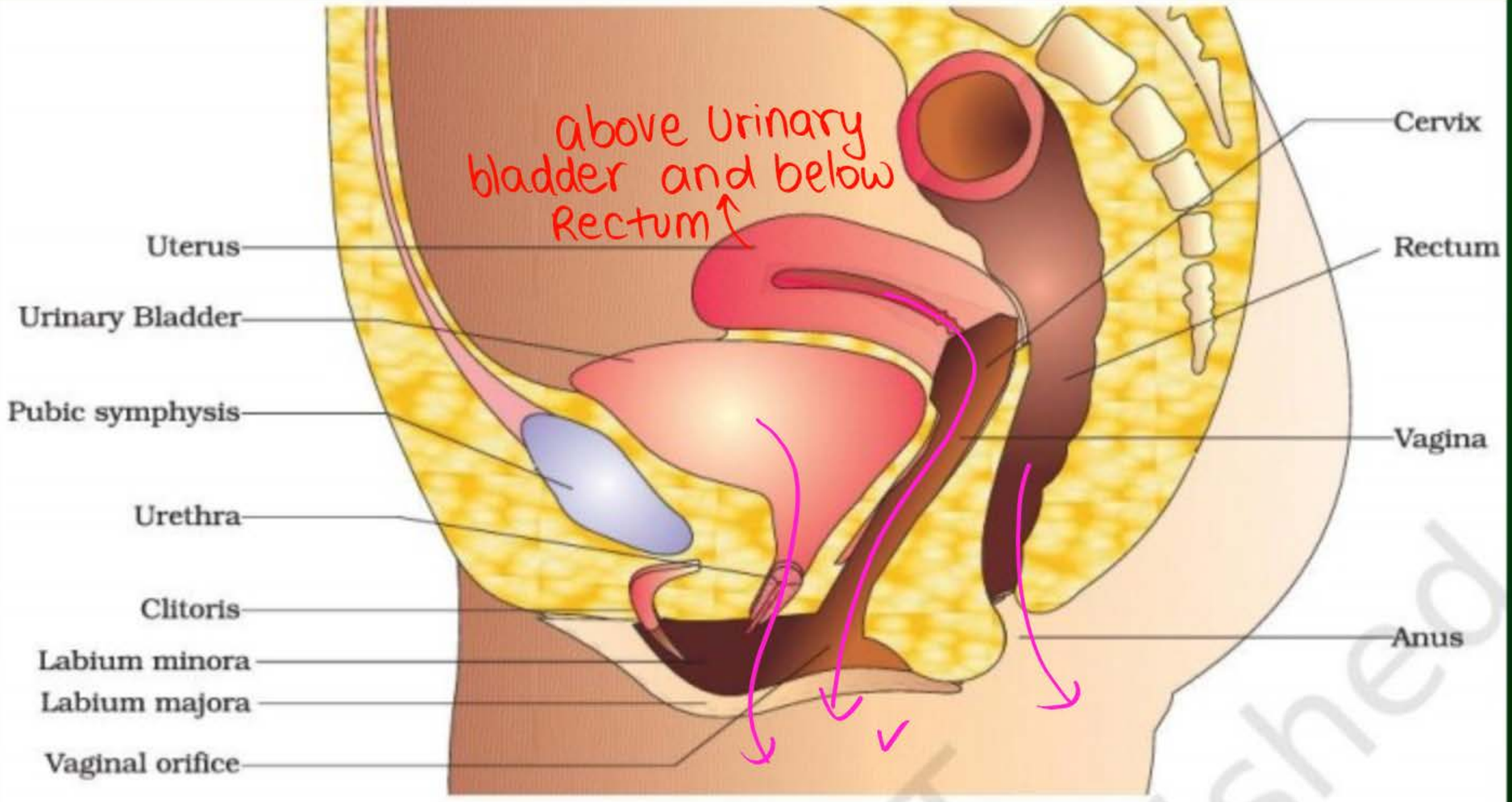


Figure 2.3 (a) Diagrammatic sectional view of female pelvis showing reproductive system

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A functional mammary gland is characteristic of all female mammals. The mammary glands are paired structures (breasts) that contain glandular tissue and variable amount of fat. The glandular tissue of each breast is divided into 15-20 **mammary lobes** containing clusters of cells called alveoli (Figure 2.4). The cells of alveoli secrete milk, which is stored in the cavities (lumens) of alveoli. The alveoli open into mammary tubules. The tubules of each lobe join to form a **mammary duct**. Several mammary ducts join to form a wider mammary ampulla which is connected to **lactiferous duct** and opens at the nipple, from where milk is expressed during breastfeeding.

Mammary gland

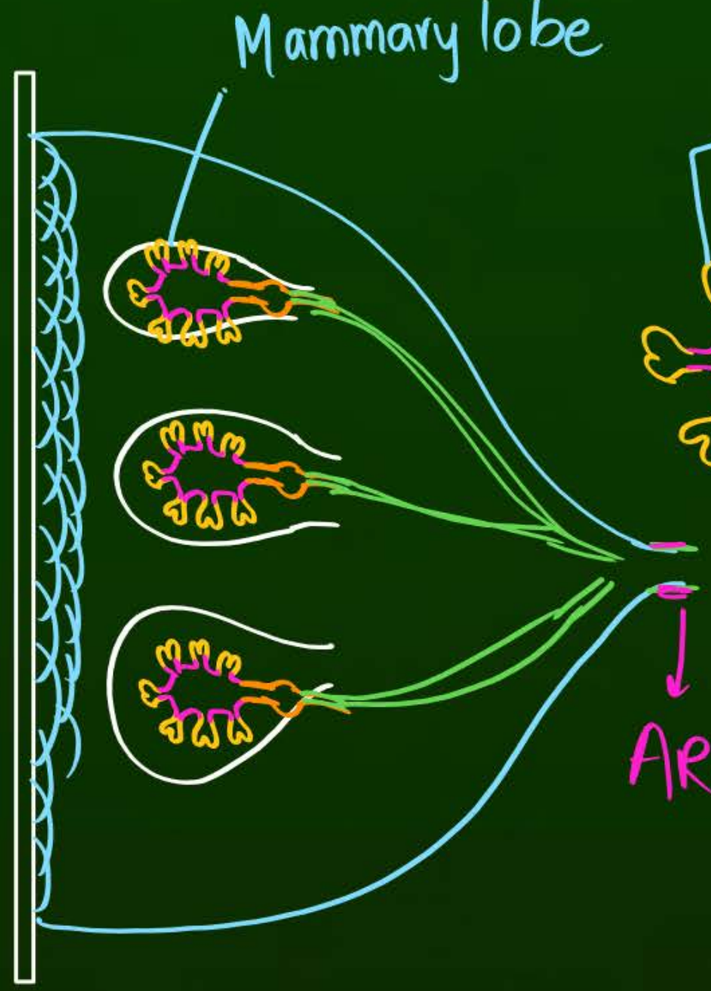
- Function mammary gland is characterstic feature of ♀ mammals
- Paired st.
- located on Pectoralis major Muscles.

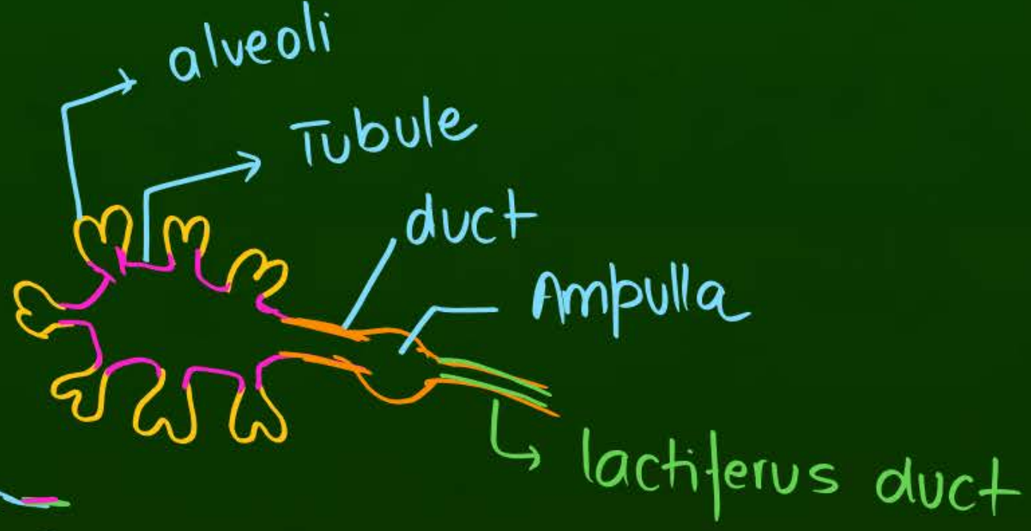
Mammary gland
- Undergo differentiation during pregnency
- Secrete milk after birth of baby.

LADDAD

- Lobe
- Aliveoli
- tubule
- Duct
- Ampulla
- Duct (lactiferous duct)
- Areola

Angry





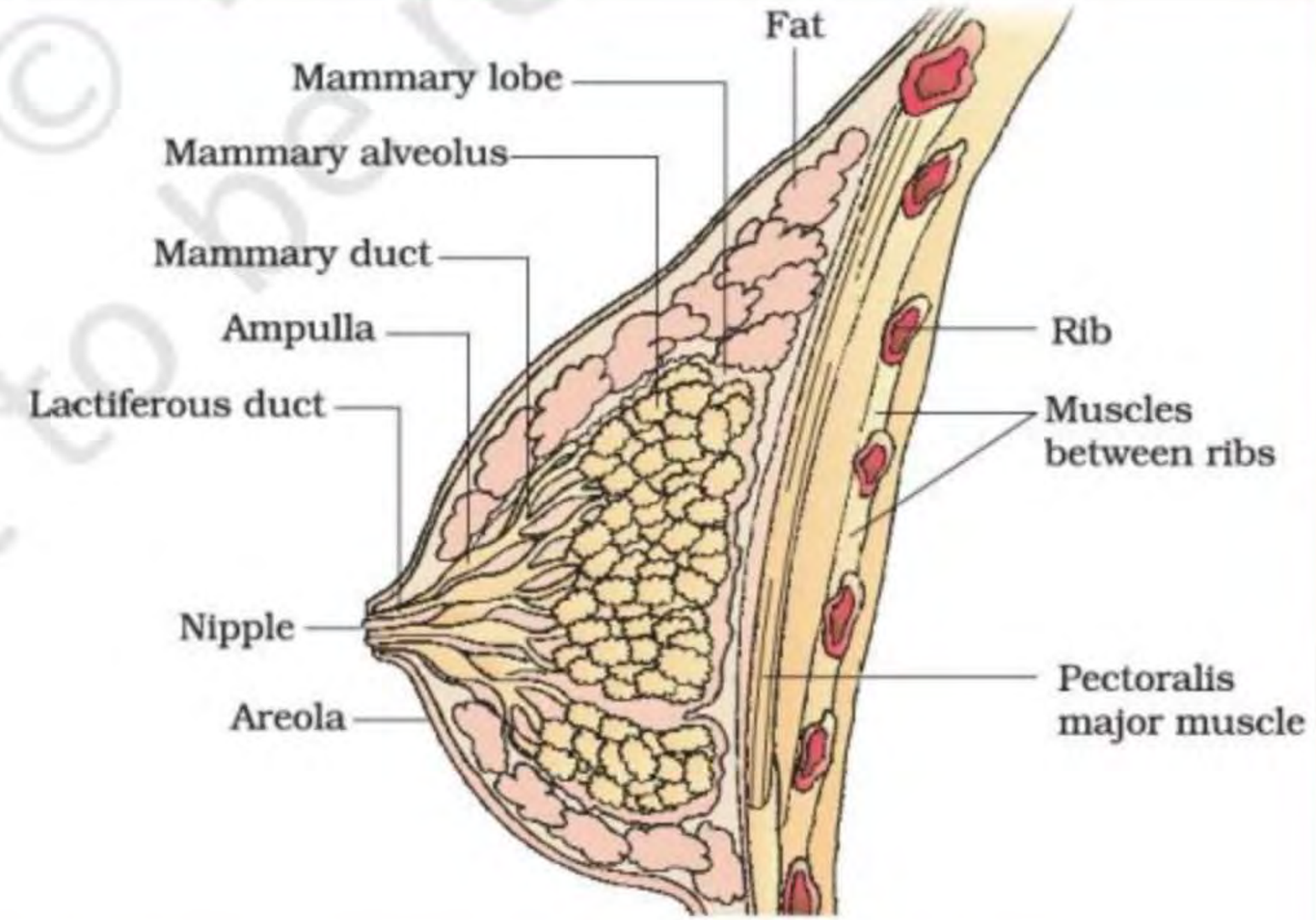


Figure 2.4 A diagrammatic sectional view of Mammary gland

QUESTION

The primary female sex organ is/are

1. Fallopian tube and cervix
2. Uterus and Infundibulum
3. Ovaries testes
4. Ovaries only



QUESTION

Which part of the oviduct remains closer to the ovary and uterus

1. Ampulla and isthmus respectively
2. Infundibulum And isthmus respectively
3. Ampulla and infundibulum respectively
4. Isthmus and infundibulum respectively



QUESTION

The uterine layer that undergoes cyclical changes during menstrual cycle is

1. Myometrium
2. Endometrium
3. Perimetrium
4. Both (a) and (c)



QUESTION

Which uterine layer exhibits strong contractions during the delivery of the baby?

1. Myometrium
2. Endometrium
3. Perimetrium
4. Both (a) and (c)



THANK YOU